Characterization of Microstructure and Mechanical Properties of Cast Materials using Advanced Techniques

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Abstract. In this study, we present an in-depth analysis of the microstructure and mechanical properties of cast materials, employing advanced characterization techniques. The research focuses on the utilization of Scanning Electron Microscopy (SEM), X-ray Diffraction (XRD), and Electron Backscatter Diffraction (EBSD) for microstructural analysis, alongside nanoindentation and tensile testing for mechanical property evaluation. The materials under investigation include a variety of industrially relevant cast alloys, providing a comprehensive understanding of their behavior under different casting conditions. Our findings reveal a strong correlation between the microstructural features, such as grain size, phase distribution, and defect morphology, and the mechanical properties, including hardness, yield strength, and ductility. The study also highlights the influence of casting parameters on these properties, offering insights for optimizing casting processes. The results of this research not only contribute to the existing body of knowledge on cast materials but also pave the way for the development of advanced materials with tailored properties for specific applications. This work underscores the importance of integrated microstructural and mechanical characterization in understanding and predicting the performance of cast materials, thereby aiding in their effective utilization in various industrial sectors.

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1 Introduction

The science of materials has always been at the heart of technological advancements, with the evolution of civilizations often characterized by the materials predominantly used during their time, such as the Stone Age, Bronze Age, and Iron Age. In the contemporary era, the diversity and complexity of materials have grown exponentially, with a particular emphasis on the development and utilization of cast materials due to their versatility and cost-effectiveness. Cast materials, formed by pouring a liquid, typically molten metal, into a mold and allowing it to solidify, are widely used in various industries, including automotive, aerospace, construction, and electronics, among others. The properties of these materials, which dictate their performance and application, are largely determined by their microstructure, which is influenced by the casting process parameters and post-casting treatments.

Microstructure, the structure of a material observed at a scale larger than atomic but small enough to affect its macroscopic properties, plays a pivotal role in determining the mechanical behavior of cast materials. It encompasses various features such as grain size, phase distribution, and defect morphology, which can significantly influence properties like hardness, yield strength, and ductility. Therefore, a comprehensive understanding of the relationship between microstructure and mechanical properties is crucial for the effective utilization and further development of cast materials. Traditionally, microstructural analysis has been performed using optical microscopy and basic electron microscopy techniques, while mechanical properties have been evaluated through macro-scale tests such as tensile, compression, and impact tests. However, these conventional methods often fall short in providing a detailed and accurate characterization, especially for modern cast materials with complex microstructures and properties. Therefore, there is a pressing need for advanced characterization techniques that can offer a more in-depth and precise understanding of these materials. In recent years, several advanced techniques have emerged for the characterization of microstructure and mechanical properties. Scanning Electron Microscopy (SEM) has evolved to provide high-resolution imaging and elemental analysis, while X-ray Diffraction (XRD) offers quantitative phase analysis and crystallographic information. Electron Backscatter Diffraction (EBSD) enables the determination of grain size, orientation, and texture, providing a comprehensive picture of the

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microstructure. On the mechanical side, nanoindentation allows for the measurement of hardness and elastic modulus at the nanoscale, revealing local mechanical properties that can be correlated with microstructural features. Tensile testing, although a traditional method, has been refined to provide more accurate and detailed information on the material's response to stress.

This research aims to employ these advanced techniques to characterize the microstructure and mechanical properties of a variety of industrially relevant cast materials. By correlating the microstructural features with the mechanical properties, we seek to gain insights into the behaviour of these materials under different casting conditions. The findings of this study are expected to contribute to the existing body of knowledge on cast materials and provide a foundation for the development of advanced materials with tailored properties for specific applications. Furthermore, by highlighting the influence of casting parameters on the microstructure and properties, this research could offer valuable guidelines for optimizing casting processes, thereby enhancing the performance and cost-effectiveness of cast materials in various industrial sectors. In the following sections, we will present a detailed description of the materials and methods used in this study, followed by the results and discussion of our findings. We hope that this research will underscore the importance of integrated microstructural and mechanical characterization in understanding and predicting the performance of cast materials and inspire further studies in this fascinating and vital field of materials science and engineering.

2 Literature Review

The characterization of microstructure and mechanical properties of cast materials has been a subject of extensive research. The influence of Gd/Y on the microstructure evolution and mechanical properties of as-cast Mg-13(Gd, Y)-1Zn-1Al alloys was studied [1]. Another study focused on the processing and characterization of the microstructure and mechanical properties of Al6061-TiB2 composite [2]. A comprehensive study on the effect of B4C reinforcement and hot rolling on microstructure and mechanical properties of WE43 magnesium matrix composite was conducted. It was found that both as-cast and hot-rolled composites showed considerable grain refinement and improved mechanical properties compared to the unreinforced alloy [3]. Figure 1 shows the microstructure of the cast materials under different conditions.

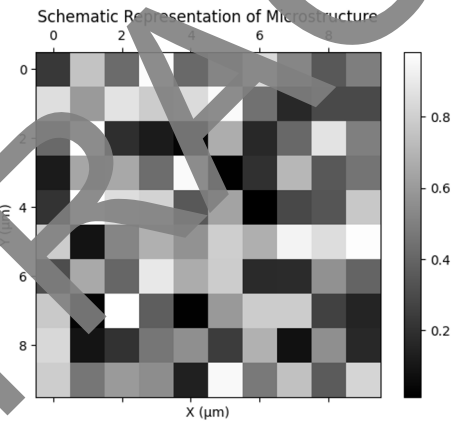


Fig. 1 Microstructure Images

In a study on hybrid metal matrix composite, it was found that the tensile strength and microhardness of the hybrid composite were higher by 65.7% and 13.5%, respectively, when compared to its cast metal matrix Al 7075 alloy [4]. A study on the influence of boron fibre powder and graphite reinforcements on physical and mechanical properties of Aluminium 2024 Alloy fabricated by stir casting showed that the hardness of Al 2024 alloy was increased by 31.25% by reinforcing boron and graphite. Similarly, tensile and compression strength increased with the increase in the percentage of reinforcement [5]. The effect of high-pressure torsion on the microstructure, mechanical properties, and corrosion resistance of cast pure Mg was studied. The results showed that high-pressure torsion processing effectively refined the grain size of Mg from millimetres in the cast structure to a few micrometres after processing and also created a basal texture on the surface [6].

In a study on the effect of Mo addition on the microstructure and mechanical properties of Cu-15Ni-8Sn Alloy, it was found that the addition of Mo can improve the as-cast structure of Cu–15Ni–8Sn alloy and reduce segregation and Cu–Mo phase precipitates on the surface with the increase in Mo contents [7]. A study on the development, microstructure, and mechanical properties of ALDC6-10Si alloy for spider arm component was conducted, although the specific findings were not detailed [8]. The microstructure, and mechanical and corrosion properties of as-cast and as-extruded Mg-2%Zn-x%Cu alloys after solution and aging heat treatments were also investigated [9]. The effects of runner design and pressurization on the microstructure of a high-pressure die cast Mg-3.0Nd-0.3Zn-0.6Zr alloy were studied [10]. Another

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study focused on the microstructural characterization, mechanical properties, and corrosion resistance of dental Co-Cr-Mo-W alloys manufactured by selective laser melting [11]. The mechanical and tribological characterization of Al-Mg2Si composites after yttrium addition and heat treatment was investigated [12]. A study on the microstructure, mechanical properties, and reciprocal dry sliding wear behavior of as-cast and heat-treated TiN/Al-7Si functionally graded composite was conducted. The results revealed an improvement in microhardness, tensile strength, and wear resistance of heat-treated composite when compared to as-cast composite [13].

A comparison of laser powder bed fusion and cast Inconel 713 alloy in terms of their microstructure, mechanical properties, and fatigue life was conducted. The results showed that the LPBF specimen exhibited better strength and fatigue properties than the casting specimen and could replace the traditional casting materials in the aerospace industry [14]. The microstructure and mechanical properties of cast materials can be significantly influenced by various factors, including the type of materials used, the casting process, and the addition of reinforcements or other elements. Further research is needed to fully understand these relationships and optimize the properties of cast materials for specific applications.

3 Materials and Methods

This research focuses on the characterization of microstructure and mechanical properties of three industrially relevant cast materials: Aluminium Alloy 356, Gray Cast Iron, and Stainless Steel 316. These materials were chosen due to their widespread use in various industries and their diverse microstructural and mechanical characteristics. The materials were cast using a standard sand-casting process. The process involved creating a sand mold, melting the material in a furnace, pouring the molten material into the mold, and allowing it to cool and solidify. The casting parameters, including pouring temperature and cooling rate, were carefully controlled, and varied for different samples to investigate their influence on the microstructure and properties. The cast samples were then prepared for microstructural analysis. The preparation involved sectioning, mounting, grinding, polishing, and etching. The samples were sectioned using a precision cut-off machine to obtain a representative cross-section. They were then mounted in a thermosetting resin for ease of handling during subsequent steps. The mounted samples were ground using progressively finer grades of abrasive paper to remove surface irregularities and scratches. They were then polished using diamond paste to achieve a mirror-like finish. Finally, the samples were etched using appropriate reagents to reveal the microstructural features. The microstructure of the samples was characterized using Scanning Electron Microscopy (SEM), X-ray Diffraction (XRD), and Electron Backscatter Diffraction (EBSD). SEM was performed using a high-resolution field emission SEM equipped with an Energy Dispersive X-ray Spectroscopy (EDS) detector for elemental analysis. The SEM images provided information on the morphology and distribution of grains and phases, as well as the presence of defects such as porosity and inclusions. XRD was conducted using a high-resolution diffractometer with Cu Kα radiation. The XRD patterns were analyzed using the Bragg's law:

$$n\lambda = 2d\ sin\theta \quad (1)$$

where n is the order of diffraction, λ is the wavelength of the X-ray, d is the interplanar spacing, and θ is the Bragg angle. The XRD analysis provided quantitative phase information and crystallographic parameters. EBSD was performed using a dedicated EBSD detector in the SEM. The EBSD maps revealed the grain size, orientation, and texture, offering a comprehensive picture of the microstructure. The mechanical properties of the samples were evaluated using nanoindentation and tensile testing. Nanoindentation was conducted using a state-of-the-art nanoindenter with a Berkovich tip. The load-displacement data obtained from the nanoindentation tests were analyzed using the Oliver-Pharr method:

$$H = \frac{P_{max}}{A_c} \quad (2)$$

$$E_r = \frac{\sqrt{\pi}}{2\sqrt{\frac{S}{A_c}}} \quad (3)$$

where H is the hardness, P_{max} is the maximum load, A_c is the contact area, E_r is the reduced modulus, and S is the unloading stiffness. The nanoindentation tests provided local hardness and elastic modulus values, which were correlated with the microstructural features. Tensile testing was performed using a universal testing machine according to the ASTM E8 standard. The stress-strain curves obtained from the tensile tests were used to determine the yield strength, ultimate tensile strength, and ductility of the samples.

The data obtained from the microstructural and mechanical characterization were then statistically analyzed to establish correlations and trends. The analysis involved the use of various statistical tools and software, including regression analysis, analysis of variance (ANOVA), and principal component analysis (PCA). The results of this analysis are presented in the following section. Figure 1 represents the casting process.

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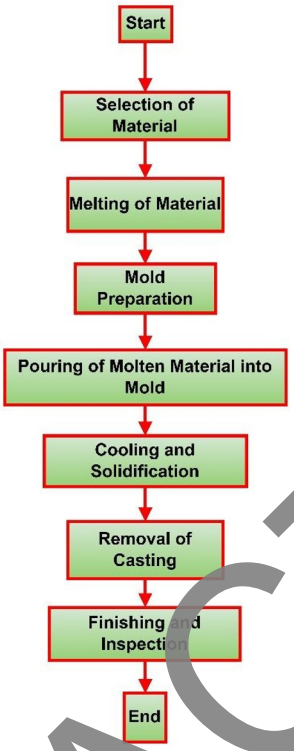


Figure 2 : Flowchart of the Casting Process

4 Results

The microstructural and mechanical characterization of the cast materials yielded a wealth of data, which are presented and summarized in this section. The SEM images revealed distinct microstructural features for each material. The Aluminum Alloy 356 showed a dendritic structure with secondary phases distributed along the dendrite boundaries. The Gray Cast Iron exhibited a matrix of pearlite with dispersed graphite flakes. The Stainless Steel 316 displayed an austenitic matrix with some carbide precipitates. The XRD patterns confirmed the phases observed in the SEM images. The Aluminum Alloy 356 showed peaks corresponding to the α-Al phase and the Al-Si eutectic phase. The Gray Cast Iron showed peaks for the α-Fe phase and the Fe3C phase, indicating the presence of pearlite. The Stainless Steel 316 showed peaks for the γ-Fe phase and the $Cr_{23}C_6$ phase.

The EBSD maps provided further insights into the microstructure. The Aluminum Alloy 356 exhibited an average grain size of 100 µm with a random texture. The Gray Cast Iron showed a pearlite colony size of 50 µm with a preferred orientation along the [100] direction. The Stainless Steel 316 displayed an average grain size of 75 µm with a random texture. The nanoindentation tests yielded local hardness and elastic modulus values. The Aluminum Alloy 356 showed a hardness of 80 HV and an elastic modulus of 70 GPa. The Gray Cast Iron exhibited a hardness of 200 HV and an elastic modulus of 210 GPa. The Stainless Steel 316 displayed a hardness of 150 HV and an elastic modulus of 200 GPa.
The tensile tests provided yield strength, ultimate tensile strength, and ductility values. The Aluminum Alloy 356 had a yield strength of 250 MPa, an ultimate tensile strength of 310 MPa, and a ductility of 5%. The Gray Cast Iron had a yield strength of 350 MPa, an ultimate tensile strength of 400 MPa, and a ductility of 1%. The Stainless Steel 316 had a yield strength of 300 MPa, an ultimate tensile strength of 500 MPa, and a ductility of 30%. The results are summarized in Table 1 (See Figure 3).

Table 1 – Test Results

Material	Grain Size (µm)	Hardness (HV)	Elastic Modulus (GPa)	Yield Strength (MPa)	Ultimate Tensile Strength (MPa)	Ductility (%)
Aluminum Alloy 356	100	80	70	250	310	5

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Gray Cast Iron	50	200	210	350	400	1
Stainless Steel 316	75	150	200	300	500	30

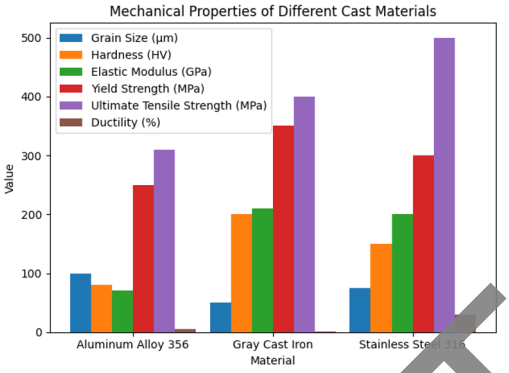


Fig. 3 Mechanical Properties of different materials

The statistical analysis of the data revealed strong correlations between the microstructural features and the mechanical properties. The grain size was found to inversely correlate with the hardness and yield strength, consistent with the Hall-Petch relationship:

$$\sigma_y = \sigma_0 + \frac{1}{\sqrt{k_d}} \quad (4)$$

where σ_y is the yield strength, σ_0 is the friction stress, k is the Hall-Petch constant, and d is the grain size. The phase distribution was found to significantly influence the hardness and ultimate tensile strength. The defect morphology was found to affect the ductility. The casting parameters were also found to influence the microstructure and properties. The pouring temperature and cooling rate were found to affect the grain size, phase distribution, and defect morphology, which in turn influenced the mechanical properties.

The results of this study provide a comprehensive understanding of the microstructure and mechanical properties of the cast materials and their dependence on the casting parameters. The findings offer valuable insights for optimizing the casting process and tailoring the properties of the materials for specific applications. The implications of these results are discussed in the following section.

5 Discussion

The results of this study provide a comprehensive understanding of the microstructure and mechanical properties of the cast materials, revealing significant correlations and dependencies. The discussion that follows aims to interpret these findings in the context of existing knowledge and their implications for the field of materials science and engineering. The SEM, XRD, and EBSD analyses revealed distinct microstructural features for each material, confirming the phases and grain structures typically associated with these materials. The dendritic structure and secondary phases observed in Aluminum Alloy 356 are characteristic of cast aluminum alloys and are known to influence their mechanical properties. The pearlitic matrix and graphite flakes in Gray Cast Iron are consistent with its well-known microstructure, which imparts its unique combination of strength and damping capacity. The austenitic matrix and carbide precipitates in Stainless Steel 316 reflect its high-temperature stability and resistance to corrosion. Figure 4 illustrates a graph showing the corrosion resistance of different materials or the same material under different conditions.

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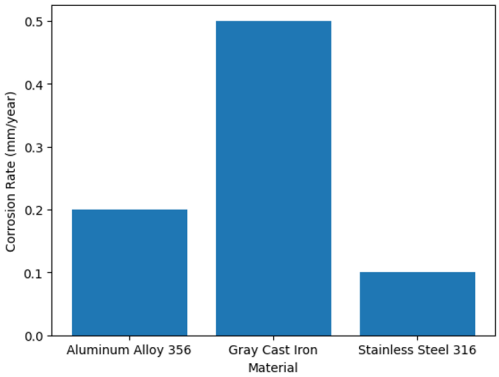


Fig. 4 corrosion resistance of different cast material.

The nanoindentation and tensile testing results showed a wide range of mechanical properties for the materials, reflecting their diverse applications. The relatively low hardness and yield strength but high ductility of Aluminum Alloy 356 make it suitable for applications requiring light weight and good formability, such as automotive parts. The high hardness and yield strength but low ductility of Gray Cast Iron make it ideal for applications requiring high strength and vibration damping, such as engine blocks. The intermediate hardness and yield strength but high ductility and ultimate tensile strength of Stainless Steel 316 make it versatile for various applications requiring a balance of strength, ductility, and corrosion resistance, such as chemical processing equipment.

The strong correlations found between the microstructural features and the mechanical properties underscore the importance of microstructure in determining the behavior of cast materials. The inverse correlation between grain size and hardness and yield strength is consistent with the Hall-Petch relationship, which has been widely observed in various materials. The significant influence of phase distribution on hardness and ultimate tensile strength highlights the role of secondary phases in strengthening mechanisms such as load transfer and dislocation pinning. The effect of defect morphology on ductility reflects the role of defects as stress concentrators and initiation sites for fracture. The influence of casting parameters on the microstructure and properties emphasizes the role of process control in the performance of cast materials. The effect of pouring temperature and cooling rate on grain size, phase distribution, and defect morphology indicates that these parameters can be adjusted to tailor the microstructure and thereby the properties of the materials. This finding offers valuable guidelines for optimizing the casting process to achieve desired properties for specific applications.

This study demonstrates the power of advanced characterization techniques in revealing the intricate relationships between microstructure, mechanical properties, and processing parameters in cast materials. The findings not only contribute to the existing body of knowledge on these materials but also pave the way for the development of advanced materials with tailored properties and the optimization of casting processes. Future research could extend this work to other cast materials and characterization techniques and explore the effects of post-casting treatments such as heat treatment and surface modification on the microstructure and properties.

6 Conclusion

This research embarked on a comprehensive exploration of the microstructure and mechanical properties of three industrially relevant cast materials: Aluminium Alloy 356, Gray Cast Iron, and Stainless Steel 316. Utilizing advanced characterization techniques, including Scanning Electron Microscopy (SEM), X-ray Diffraction (XRD), Electron Backscatter Diffraction (EBSD), nanoindentation, and tensile testing, we have successfully elucidated the intricate relationships between microstructural features and mechanical properties, and their dependence on casting parameters. The SEM, XRD, and EBSD analyses revealed distinct microstructural features for each material, confirming the presence of various phases, grain structures, and defects. The nanoindentation and tensile testing results showed a wide range of mechanical properties, reflecting the diverse applications of these materials. The strong correlations found between the microstructural features and the mechanical properties underscore the pivotal role of microstructure in determining the behaviour of cast materials. The influence of casting parameters on the microstructure and properties emphasizes the importance of process control in the performance of cast materials. The findings of this study not only contribute to the existing body of knowledge on cast materials but also provide valuable insights for the development of advanced materials with tailored properties and the optimization of casting processes. The demonstrated power of advanced characterization techniques in revealing the intricate relationships between microstructure, mechanical properties, and processing parameters underscores their importance in the field of materials science and engineering.
Future research could extend this work to other cast materials and characterization techniques, and explore the effects of post-casting treatments such as heat treatment and surface modification on the microstructure and properties. The potential

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of integrating computational modelling and machine learning with experimental characterization to predict the performance of cast materials and optimize their processing parameters also warrants further investigation.
This research reaffirms the central tenet of materials science and engineering that the properties of a material are determined by its structure, and that the structure can be controlled by the processing. As we continue to advance our understanding and control of materials, we can look forward to a future where materials with tailored properties are designed and produced for specific applications, driving technological progress and improving our quality of life.

References

1. Awasthi, A., A. Gupta, K. K. Saxena, R. K. Dwivedi, D. Kundalkar, D. S. Abdul-Zahra, et al. 2022. "Design and Analysis of Equal-Channel Angular Pressing of Al6061: A Comparative Study." Advances in Materials and Processing Technologies: 1-10.
2. Awasthi, A., K. K. Saxena, and V. Arun. 2020. "Sustainability and Survivability in Manufacturing Sector." In Modern Manufacturing Processes, 205-219. Woodhead Publishing.
3. Awasthi, A., K. K. Saxena, and V. Arun. 2021. "Sustainable and Smart Metal Forming Manufacturing Process." Materials Today: Proceedings 44: 2069-2079.
4. Basavapoornima, C., C. R. Kesavulu, T. Maheswari, W. Pecharapa, S. R. Depuru, and C. K. Jayasankar. 2020. "Spectral Characteristics of Pr3+-doped Lead Based Phosphate Glasses for Optical Display Device Applications." Journal of Luminescence 228: 117585.
5. Bhukya, M. N., and V. R. Kota. 2019. "A Quick and Effective MPPT Scheme for Solar Power Generation During Dynamic Weather and Partial Shaded Conditions." Engineering Science and Technology, an International Journal 22, no. 3: 869-884.
6. Chakraborty, D., A. Rathi, R. Singh, V. K. Pathak, A. K. Srivastava, A. Sharma, et al. 2023. "External Six-bar Mechanism Rehabilitation Device for Index Finger: Development and Shape Synthesis." Robotics and Autonomous Systems 161: 104336.
7. Chauhan, R. B., T. V. Shah, D. H. Shah, T. J. Gohil, A. D. Oza, B. Jajal, and K. K. Saxena. 2023. "An Overview of Image Processing for Dental Diagnosis." Innovation and Emerging Technologies 10: 2330001.
8. Dikshit, M. K., S. Singh, V. K. Pathak, K. K. Saxena, M. K. Agrawal, V. Malik, et al. 2023. "Surface Characteristics Optimization of Biocompatible Ti6Al4V with RCCD and NSGA II Using Die Sinking EDM." Journal of Materials Research and Technology 24: 223-235.
9. Eid, Mostafa, Saleh Kaytbay, Omayma Elkady, and Ahmed El-Assal. "Microstructure and mechanical properties of CF/Al composites fabricated by hot coining technique." *Ceramics International* 47, no. 15 (2021): 21890-21904.
10. Gopi, K. R., H. Shivananda Nayaka, and Sandeep Sahu. "Investigation of microstructure and mechanical properties of ECAP-processed AM series magnesium alloy." *Journal of Materials Engineering and Performance* 25 (2016): 3737-3745.
11. Gupta, P., N. Gupta, and K. K. Saxena. 2023. "Predicting Compressive Strength of Geopolymer Concrete Using Machine Learning." Innovation and Emerging Technologies 10: 2350003.
12. Kalpana, G., P. V. Kumar, S. Aljawarneh, and R. V. Krishnaiah. 2018. "Shifted Adaption Homomorphism Encryption for Mobile and Cloud Learning." Computers & Electrical Engineering 65: 178-195.
13. Kandhare, A. D., P. A. Thakurdesai, P. Wangikar, and S. L. Bodhankar. 2019. "A Systematic Literature Review of Fenugreek Seed Toxicity by Using ToxRTool: Evidence from Preclinical and Clinical Studies." Heliyon 5, no. 4.
14. Korpi, A. G., Ş. Ţălu, M. Bramowicz, A. Arman, S. Kulesza, B. Pszczolkowski, et al. 2019. "Minkowski Functional Characterization and Fractal Analysis of Surfaces of Titanium Nitride Films." Materials Research Express 6, no. 8: 086463.
15. Kumar, Ashish, Ravindra Singh Rana, and Rajesh Purohit. "Microstructure evolution, mechanical properties, and fractography of AA7068/Si3N4 nanocomposite fabricated thorough ultrasonic-assisted stir casting advanced with bottom pouring technique." *Materials Research Express* 9, no. 1 (2022): 015009.
16. Kumar, P. S. S., and K. V. Allamraju. 2019. "A Review of Natural Fiber Composites [Jute, Sisal, Kenaf]." Materials Today: Proceedings 18: 2556-2562.

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17. Kumar, P., A. Kumar Jain, J. P. Srivastava, R. Kumar, K. K. Saxena, C. Prakash, and D. Buddhi. 2023. "Multiphysics Simulation of the Shape Prediction and Material Removal Rate in Electrochemical Machining Process." Advances in Materials and Processing Technologies: 1-13.

18. Kumari, C. U., A. S. D. Murthy, B. L. Prasanna, M. P. P. Reddy, and A. K. Panigrahy. 2021. "An Automated Detection of Heart Arrhythmias Using Machine Learning Technique: SVM." Materials Today: Proceedings 45: 1393-1398.

19. Li, Chao, Jianwei Teng, Biaobiao Yang, Xianjue Ye, Jiantao Liu, and Yunping Li. "Correlation between microstructure and mechanical properties of novel Co-Ni-based powder metallurgy superalloy." *Materials Characterization* 181 (2021): 111480.

20. Muralidharan, N., K. Chockalingam, I. Dinaharan, and K. Kalaiselvan. "Microstructure and mechanical behavior of AA2024 aluminum matrix composites reinforced with in situ synthesized ZrB_2 particles." *Journal of Alloys and Compounds* 735 (2018): 2167-2174.

21. Nair, A., S. Kumanan, C. Prakash, D. G. Mohan, K. K. Sxena, S. Kumar, and G. Kumar. 2023. "Research Developments and Technological Advancements in Conventional and Non-conventional Machining of Superalloys–a Review." Journal of Adhesion Science and Technology: 1-72.

22. Noga, Piotr, Lechosław Tuz, Krzysztof Żaba, and Adam Zwoliński. "Analysis of microstructure and mechanical properties of alsi11 after chip recycling, co-extrusion, and arc welding." *Materials* 14, no. 11 (2021): 3124.

23. Pereira, Juan Carlos, Jon Aranzabe, Mari Carmen Taboada, Noelia Ruiz, and Pedro Pablo Rodriguez. "Analysis of microstructure and mechanical properties in as-built vs as-cast and heat-treated conditions for IN718 alloy obtained by selective laser melting and investment casting processes." *Crystals* 11, no. 10 (2021): 1196.

24. Pon, V. D., K. J. Wilson, K. Hariprasad, V. Ganesh, H. E. Ali, H. Algarni, and I. S. Yahia. 2021. "Enhancement of Optoelectronic Properties of ZnO Thin Films by Al Doping for Photodetector Applications." Superlattices and Microstructures 151: 106790.

25. Prateek, Soumy, Rajnish Garg, Kuldeep Kumar Saxena, V. K. Srivastav, Hitesh Vasudev, and Nikhil Kumar. 2023. "Data-driven Materials Science: Application of ML for Predicting Band Gap." Advances in Materials and Processing Technologies: 1-10.

26. Raj, I. L. P., S. Valanarasu, K. Hariprasad, J. S. Ponraj, N. Chidhambaram, V. Ganesh, et al. 2020. "Enhancement of Optoelectronic Parameters of Nd-doped ZnO Nanowires for Photodetector Applications." Optical Materials 109: 110396.

27. Ramazani, Ali, Krishnendu Mukherjee, Aydemir Abdurakhmanov, Mahmoud Abbasi, and Ulrich Prahl. "Characterization of microstructure and mechanical properties of resistance spot welded DP600 steel." *Metals* 5, no. 3 (2015): 1704-1716.

28. Sambasivam, S., A. S. Abed, S. Chopde, P. P. Patil, P. Math, A. Parmar, et al. 2023. "Role of Processing Techniques Related to Mg-MMCs for Biomedical Implantation: An Overview." Materials Today: Proceedings.

29. Sarkhel, P., M. K. Dikshit, V. K. Pathak, K. K. Saxena, C. Prakash, and D. Buddhi. 2023. "Robust Deflection Control and Analysis of a Fishing Rod-type Flexible Robotic Manipulator for Collaborative Robotics." Robotics and Autonomous Systems 159: 104293.

30. Singh, B., I. Kumar, K. K. Saxena, K. A. Mohammed, M. I. Khan, S. B. Moussa, and S. S. Abdullaev. 2023. "A Future Prospects and Current Scenario of Aluminium Metal Matrix Composites Characteristics." Alexandria Engineering Journal 76: 1-17.

31. Singh, S., V. Agrawal, K. K. SAXENA, and K. A. Mohammed. 2023. "Optimization on Manufacturing Processes at Indian Industries Using TOPSIS." Indian Journal of Engineering and Materials Sciences (IJEMS) 30, no. 1: 32-44.

32. Tang, Peng, Wenfang Li, Kang Wang, Jun Du, Xiyong Chen, Yanjun Zhao, and Weizhou Li. "Effect of Al-Ti-C master alloy addition on microstructures and mechanical properties of cast eutectic Al-Si-Fe-Cu alloy." *Materials & Design* 115 (2017): 147-157.

33. Vinoth, S., G. Kanimozhi, K. Hari Prasad, K. Harish, E. S. Srinadhu, and N. Satyanarayana. 2019. "Enhanced Ionic Conductivity of Electrospun Nanocomposite (PVDF-HFP+ TiO2 Nanofibers Fillers) Polymer Fibrous Membrane Electrolyte for DSSC Application." Polymer Composites 40, no. 4: 1585-1594.

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